ARCHITECTURE 1

DD + DeepSets + conditional MAF

Registered family: dd_deepsets_maf | current production training path | PARTONS remains the sole forward-physics backend

A. Observation encoder (permutation invariant)

D Dataset context

36 tokens = 6 kinematics x 6 observables
18 ordered features/token
+ 9 global features
flattened width = 657

Shared point MLP h_psi

Linear 18->128; SiLU
LayerNorm: disabled (Identity)
Linear 128->128; SiLU
Linear 128->128; SiLU
Same weights for all 36 tokens

Mask + symmetric pool

e_i = h_psi(x_i) * point_mask_i
all active: mean(e_i)
masked: sum(e_i) / 36
then concatenate 9 globals
width: 128 + 9 = 137

Dataset MLP g_rho

Linear 137->128; SiLU
Linear 128->96
(no final activation)
DeepSets parameters: 65,504

Embedding e_D

shape [batch, 96]
Conditions every flow transform

Conditional MAF

q_phi(z | e_D)
82-dimensional
6 affine autoregressive transforms

B. Exact target transform and density estimator

condition e_D

Physical posterior target theta

80 DD controls = 4 GPDs x 4 channels x 5 fields
fields: normalization, a, c, profile_b, t slope
+ 2 standard-normal normalization nuisances

Analytic prior transform

Each bounded DD coordinate:
z = Phi^-1((theta - lower)/width)
Two nuisance coordinates: identity
Result: z in R^82 with N(0, I) prior

sbi preprocessing

z_score_theta = independent
z_score_x = none
(Context already explicitly scaled.)

Six-transform Zuko affine MAF

Base: u ~ N(0, I_82)
Orders alternate forward / reversed (randperm=False)
Each masked conditioner sees earlier coordinates + e_D
6 ReLU hidden layers, width 96; final width 164
= shift + scale for each of 82 coordinates

Posterior samples

inverse estimator z-score
inverse probit -> theta
q_phi(theta | D)
82 physical coordinates

exact PARTONS

Frozen input contract

18 point features (ordered): x_B/0.3; -t/0.2; Q2/Q0^2; E_beam/12; sin(phi); cos(phi); six observable one-hot bits; observed value; marginal sigma; symmetric C^(-1/2)y/10; global-normalization response; LU-normalization response; point_mask. Nine globals (ordered): PARTONS-backend flag; Q0^2/(1 GeV^2); LO-coefficient flag; configured-evolution flag; native datum-Q2 evolution flag; full-covariance flag; four-twist-2-GPD flag; native-six-observable-route flag; model_mask.

Interpretation and boundary

The network learns a normalized joint posterior, not 82 independent regressions. Dashed downstream physics is intentionally outside the neural network: posterior samples are reevaluated by the exact native PARTONS bridge for CFFs, observables, and GPD curves.

ARCHITECTURE 2

NNGPD-inspired latent function + DeepSets + conditional MAF

Registered family: neural_gpd_deepsets_maf | function autoencoder v1 | latent-function closure representation

A. Function autoencoder (trained only from stored canonical GPD truth)

Function tokens

For each canonical point i:
coordinate c_i in R^d_c
scalar truth value f_i
validity mask m_i
token width = d_c + 1

Shared function encoder

Linear (d_c+1)->128; Tanh
Linear 128->128; Tanh
multiply output by m_i
active mean: sum / clamp(sum m_i, 1)
Linear 128->16

Latent code ell

shape [group, 16]
Default latent_dim = 16

Coordinate-conditioned decoder

Expand ell over coordinates; concatenate [c_i, ell]
Linear (d_c+16)->128; Tanh
Linear 128->128; Tanh
Linear 128->128; Tanh
Linear 128->1

Reconstruction

f_hat_i for every stored coordinate
masked mean-squared error

Selection

Adam, lr 1e-3
100 epochs
best validation MSE
seed 73129

ell targets

Train-only latent standardization

diagonal Gaussian mean/scale fitted on training ell only

B. Amortized posterior over function latents

Observation context D

Current release contract:
36 x 18 point-token tensor
+ 9 globals (width 657)
No truth fields enter context

DeepSets encoder

Point MLP: 18->128->128->128, SiLU
post-network mask; fixed-maximum pooling
concat 9 globals
Set MLP: 137->128->96, SiLU between
output e_D in R^96

Dataset embedding e_D

shape [batch, 96]
Permutation invariant
Conditions the posterior density; it is not a GPD-function latent

Conditional MAF family

sbi.NPE + zuko_maf
conditioned on e_D
models joint posterior over standardized function latent
MAF is the only registered density estimator

Latent draws

ell ~ q_phi(ell | D)
unstandardize with fitted diagonal prior

Decode

[c_i, ell]
through frozen decoder
-> GPD values

Exact scope and safeguards

Identifier: permutation_invariant_function_autoencoder_v1; decoder: coordinate_conditioned_mlp_v1. Default d_c is artifact-defined (the checkpoint records coordinate_width), so it is intentionally symbolic here. The support guard rejects unsupported coordinates, and the published decoder contract uses extrapolation_policy='forbidden'. Its fitted latent prior uses training groups only, and train / validation / locked-holdout roles must be nonempty and disjoint.

Implementation status (current worktree)

The family and function autoencoder/model-view publisher are implemented and registered. The ordinary workflows/pseudodata.py train_model() path still loads theta_latent and writes a dd_deepsets_maf result bundle. Therefore this page describes the registered neural-GPD architecture without claiming that the standard CLI currently trains it end-to-end. The neural model-view publisher does not yet record concrete MAF widths/transform count.